LuSEE-Night Technical Memo

Document: LSN-SPT-0110

Title: Measurement of the input impedance of the LuSEE-Night preamplifier (IO-1892)

Revision: 00

Revision Date: 05/27/2025

Reference Drawing:

Purpose/Background

This document describes how to measure the input impedance of the LuSEE-Night preamplifier board (IO-1892).

Note: Variations of this procedure are permitted on a case by case basis, so as a robust and safe measurement is achieved.

Tools and materials required:

- Antistatic wrist strap
- Rohde & Schwarz LCR meter, LCX200, with Kelvin clip (1 set)
- SMA 50-ohm terminator (2 pcs)
- SMA female-female adapter (1 pcs)
- SMA edge-mount female connector (1 pcs)

Procedure:

1) Wear an antistatic wrist strap or an equivalent.
2) Allow the LCR200 to power on for at least 30 minutes before taking a measurement.

3) Set the LCR200 to measure impedance by pressing Functions -> “R” if it is not already.
4) Set the bias voltage to “0.1 Vrms” if it is not already.
5) Set the range to “Man: 3kΩ” if it is not already.

6) Disconnect the coaxial cable (C0 hereaeer) connecting the preamplifier J3 to the spectrometer's motherboard, but only at the preamplifier end. Then, connect a 50-ohm SMA terminator with the SMA female-female adapter to that end of the cable. [SKIP THIS WHOLE STEP]

7) Connect the 50-ohm SMA terminator to the J3 connector on the preamplifier as shown in Fig. 1. [SKIP THIS STEP]

8) Power on the LuSEE-Night and do the appropriate system check (follow the LuSEE-Night system procedure).

9) Power on the analog power of the spectrometer and do the health check (follow the LuSEE-Night system procedure).

10) Disconnect the terminator connected to preamplifier J2 .

11) Connect the edge-mount SMA female connector to the J2 connector on the preamplifier as shown in Fig. 1.

12) Use the red Kelvin clip on the LCX200 to connect to the central conductor of the edge-mount SMA connector and use the black Kelvin clip to connect to one of the ground pins of the SMA connector as shown in Fig. 2. Make sure not to create a short circuit between the red and black Kelvin clips.

13) Set the LCX200's frequency to 100 kHz, then press “Trig.” to measure the impedance. When “RANGEΩ” appears in Z measurement, continue pressing “Trig.” until a numerical value appears as shown in Fig. 3.
14) Take the measurement at least five times to calculate the standard deviation.
15) Repeat steps 13) and 14) for the following frequencies: 180 kHz, 320 kHz, 560 kHz, and 1 MHz. Figure 4 shows the input impedance of FM preamplifier #1 with the long antenna input coaxial cable measured at BNL as an example.

16) Disconnect the Kelvin clips and the edge-mount SMA connector from the preamplifier J2 connector.

17) Use the torque wrench to connect the preamplifier J2 connector to the antenna with the appropriate torque (follow the LuSEE-Night system procedure).

18) Turn off the analog power of the spectrometer and do the health check (follow the LuSEE-Night system procedure).

19) Turn off the LuSEE-Night and do the appropriate system check (follow the LuSEE-Night system procedure).

20) Disconnect the 50-ohm SMA terminator from the preamplifier J3 connector.

21) Disconnect the 50-ohm SMA terminator with the SMA female-female adapter from C0 and connect C0 to the preamplifier J3 connector with the appropriate torque (follow the LuSEE-Night system procedure).

22) Repeat steps 6) to 21) for other preamplifiers.

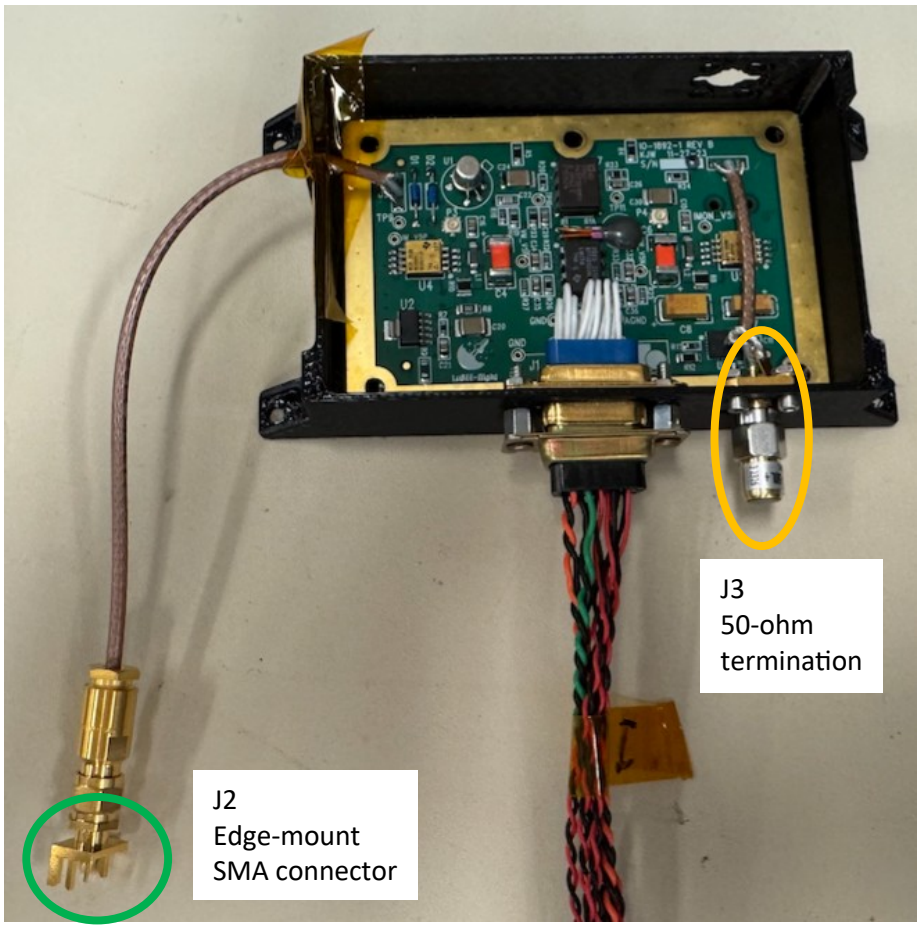


Fig. 1 J2 and J3 of the FM preamplifier connection.

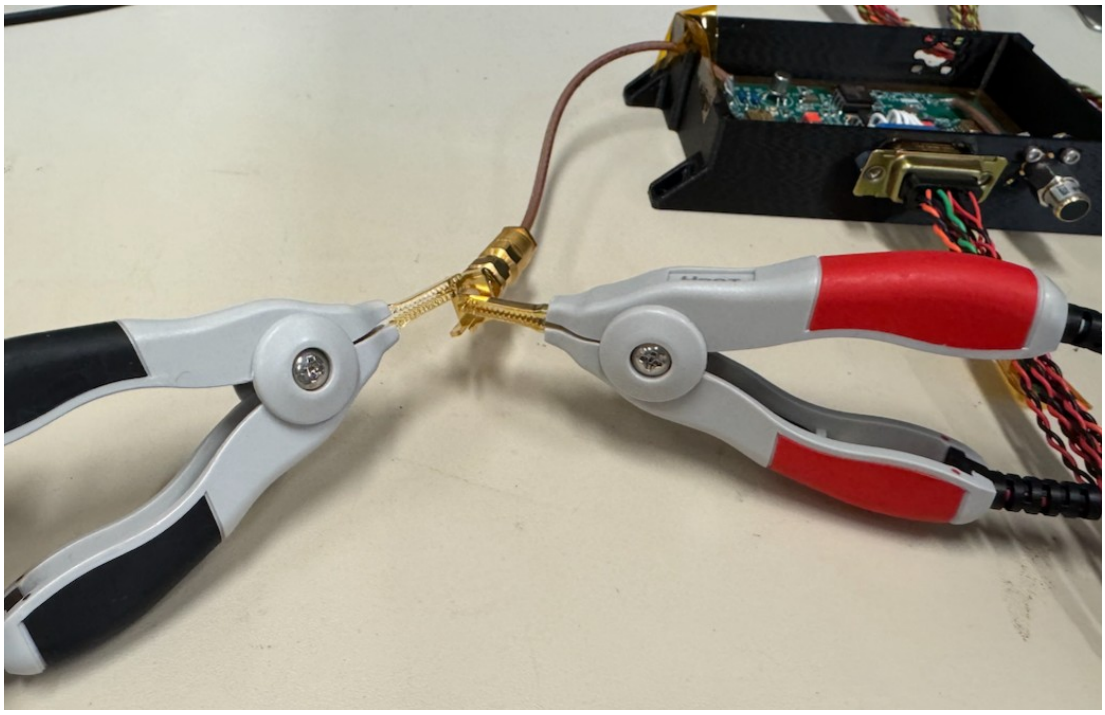


Fig. 2 Kelvin clip connecLon to the SMA edge-mount connector on J2. Verify not to create a short circuit between the red and black Kelvin clips.



Fig. 3 Front panel of the LCX200 when measured at 500 kHz. Verify that the$\theta_{\text{¦}}$ value is approximately -90°, indicaLng purely capaciLve impedance.

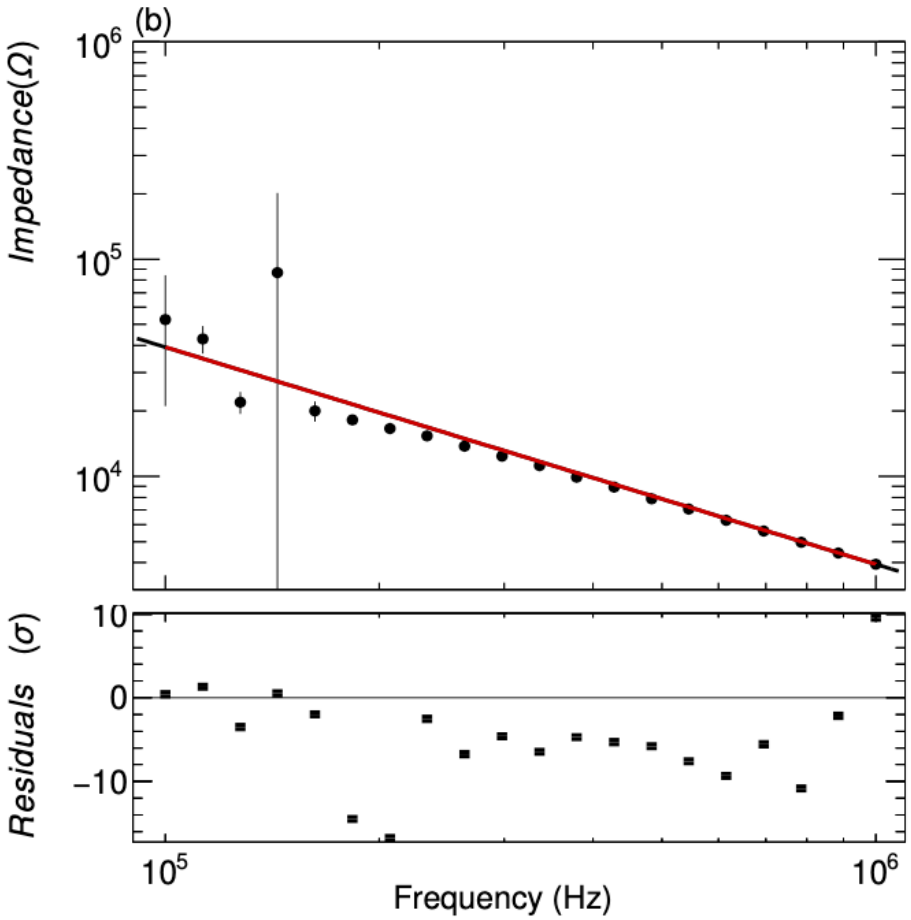


Fig. 4 Input impedance of the FM preamplifier #1 measured at BNL with the long input coaxial cable ("Characterization of the flight model preamplifier for LuSEE-Night", 2024, BNL).